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Parametric and Non-Parametric Tests of Independence

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TESTS CONCERNING CORRELATION

Learning Outcome

- explain parametric and nonparametric tests of the hypothesis that the population correlation coefficient equals zero, and determine whether the hypothesis is rejected at a given level of significance

The most common hypotheses concerning correlation occur when comparing the population correlation coefficient with zero because we often ask whether a relationship exists, which implies a null of the correlation coefficient equal to zero (i.e., no relationship). Hypotheses concerning the population correlation coefficient may be two- or one-sided, as we have seen in other tests. Let ρ represent the population correlation coefficient. The possible hypotheses are as follows:

Two sided: H_0: $\rho = 0$ versus H_a: $\rho \neq 0$

One sided (right side): H_0: $\rho \leq 0$ versus H_a: $\rho > 0$

One sided (left side): H_0: $\rho \geq 0$ versus H_a: $\rho < 0$

We use the sample correlation to test these hypotheses on the population correlation.

Parametric Test of a Correlation

The parametric pairwise correlation coefficient is often referred to as **Pearson correlation**, the **bivariate correlation**, or simply the correlation. Our focus is on the testing of the correlation and not the actual calculation of this statistic, but it helps distinguish this correlation from the nonparametric correlation if we look at the formula for the sample correlation. Consider two variables, X and Y. The sample correlation, r_{XY}, is as follows:

$$r_{XY} = \frac{s_{XY}}{s_X s_Y}, \qquad 1$$

where s_{XY} is the sample covariance between the X and Y variables, s_X is the standard deviation of the X variable, and s_Y is the standard deviation of the Y variable. We often drop the subscript to represent the correlation as simply r.

Therefore, you can see from this formula that each observation is compared with its respective variable mean and that, because of the covariance, it matters how much each observation differs from its respective variable mean. Note that the covariance drives the sign of the correlation.

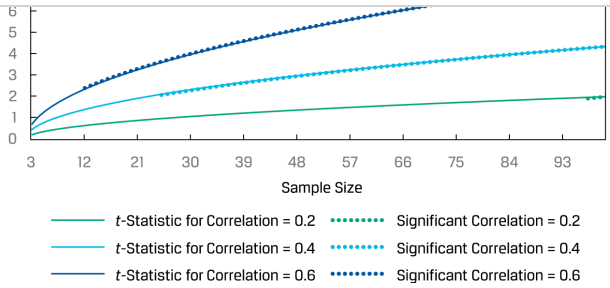
If the two variables are normally distributed, we can test to determine whether the null hypothesis (H_0: $\rho = 0$) should be rejected using the sample correlation, r. The formula for the t-test is as follows:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}. \qquad 2$$

This test statistic is t-distributed with $n - 2$ degrees of freedom. One practical observation concerning Equation 2 is that the magnitude of r needed to reject the null hypothesis decreases as sample size n increases, for two reasons. First, as n increases, the number of degrees of freedom increases and the absolute value of the critical value of the t-statistic decreases. Second, the absolute value of the numerator increases with larger n, resulting in a larger magnitude of the calculated t-statistic. For example, with sample size $n = 12$, $r = 0.35$ results in a t-statistic of 1.182, which is not different from zero at the 0.05 level ($t_{\alpha/2} = \pm 2.228$). With a sample size of $n = 32$, the same sample correlation, $r = 0.35$, yields a t-statistic of 2.046, which is just significant at the 0.05 level ($t_{\alpha/2} = \pm 2.042$).

Another way to make this point is that when sampling from the same population, a false null hypothesis is more likely to be rejected (i.e., the power of the test increases) as we increase the sample size, all else equal, because a higher number of observations increases the numerator of the test statistic. We show this in Exhibit 1 for three different sample correlation coefficients, with the corresponding calculated t-statistics and significance at the 5 percent level for a two-sided alternative hypothesis. As the sample size increases, significance is more likely to be indicated, but the rate of achieving this significance depends on the sample correlation coefficient; the higher the sample correlation, the faster significance is achieved when increasing the sample size. As the sample sizes increase as ever-larger datasets are examined, the null hypothesis is almost always rejected and other tools of data analysis must be applied.

Exhibit 1: Calculated Test Statistics for Different Sample Sizes and Sample Correlations with a 5 Percent Level of Significance



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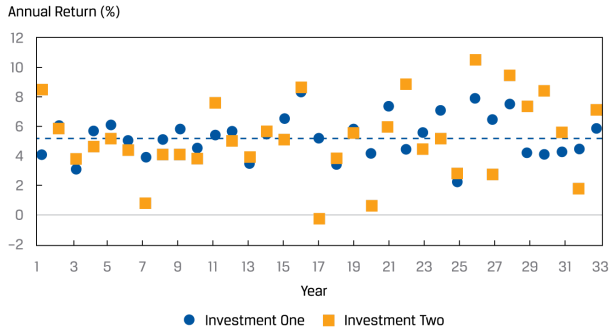
EXAMPLE 1

Examining the Relationship between Returns on Investment One and Investment Two

An analyst is examining the annual returns for Investment One and Investment Two over 33 years, as displayed in Exhibit 2.

Exhibit 2: Returns for Investments One and Two over 33 Years



Although this time series plot provides some useful information, the analyst is most interested in quantifying how the returns of these two series are related, so she calculates the correlation coefficient, equal to 0.43051, between these series.

1. Is there a significant positive correlation between these two return series if she uses a 1 percent level of significance?

Hide Solution

Solution:

Step 1	State the hypotheses.	$H_0 : \rho \leq 0$ versus $H_a : \rho > 0$
Step 2	Identify the appropriate test statistic.	$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$
Step 3	Specify the level of significance.	1%
Step 4	State the decision rule.	With 33 − 2 = 31 degrees of freedom and a one-sided test with a 1% level of significance, the critical value is 2.45282. We reject the null hypothesis if the calculated *t*-statistic is greater than 2.45282.
Step 5	Calculate the test statistic.	$t = \frac{0.43051\sqrt{33-2}}{\sqrt{1-0.18534}}$ $= 2.65568$
Step 6	Make a decision.	Reject the null hypothesis because the calculated *t*-statistic is greater than 2.45282. Evidence is sufficient to reject the H_0 in favor of H_a, that the correlation between the annual returns of these two investments is positive.

EXAMPLE 2

Correlation with the S&P 500 Returns

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Exhibit 3: Sample Correlations between Monthly Returns and the S&P 500

Fund 1	Large-cap fund
Fund 2	Mid-cap fund
Fund 3	Large-cap value fund
Fund 4	Emerging market fund
S&P 500	US domestic stock index

	Fund 1	Fund 2	Fund 3	Fund 4	S&P 500
Fund 1	1				
Fund 2	0.9231	1			
Fund 3	0.4771	0.4156	1		
Fund 4	0.7111	0.7238	0.3102	1	
S&P 500	0.8277	0.8223	0.5791	0.7515	1

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Test the null hypothesis that each of these correlations, individually, is equal to zero against the alternative hypothesis that it is not equal to zero. Use a 5 percent significance level and critical *t*-values of ±2.032.

Hide Solution

Solution:

The hypotheses are H_0: $\rho = 0$ and H_a: $\rho \neq 0$. The calculated test statistics are based on the formula

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}.$$

For example, the calculated *t*-statistic for the correlation of Fund 3 and Fund 4 is as follows

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.3102\sqrt{36-2}}{\sqrt{1-0.3102^2}} = 1.903.$$

Repeating this calculation for the entire matrix of correlations gives the following:

Calculated *t*-Statistics for Correlations

	Fund 1	Fund 2	Fund 3	Fund 4	S&P 500
Fund 1					
Fund 2	13.997				
Fund 3	3.165	2.664			
Fund 4	5.897	6.116	1.903		
S&P 500	8.600	8.426	4.142	6.642	

With critical values of ±2.032, with the exception of the correlation between Fund 3 and Fund 4 returns, we reject the null hypothesis for these correlations. In other words, evidence is sufficient to indicate that the correlations are different from zero, with the exception of the correlation of returns between Fund 3 and Fund 4.

We could use software to determine the critical values:

Excel

```
[T.INV(0.025,34) AND T.INV(0.975,34)]
```

R

```
[QT(C(.025,.975),34)]
```

Python

```
[FROM SCIPY.STATS IMPORT T AND T.PPF(.025,34) AND T.PPF(.975,34)]
```

We also could use software to determine the *p*-value for the calculated test statistic to enable a comparison with the level of significance. For example, for $t = 2.664$, the *p*-value is 0.01172:

Excel

```
[(1-T.DIST(2.664,34,TRUE))*2]
```

R

```
(1-T.CDF(2.664,34))*2]
```

Non-Parametric Test of Correlation: The Spearman Rank Correlation Coefficient

When we believe that the population under consideration meaningfully departs from normality, we can use a test based on the **Spearman rank correlation coefficient**, r_S. The Spearman rank correlation coefficient is essentially equivalent to the Pearson correlation coefficient as defined earlier, but it is calculated on the *ranks* of the two variables (e.g., *X* and *Y*) within their respective samples. The calculation of r_S requires the following steps:

1. Rank the observations on *X* from largest to smallest. Assign the number 1 to the observation with the largest value, the number 2 to the observation with second largest value, and so on. In case of ties, assign to each tied observation the average of the ranks that they jointly occupy. For example, if the third and fourth largest values are tied, we assign both observations the rank of 3.5 (the average of 3 and 4). Perform the same procedure for the observations on *Y*.
2. Calculate the difference, d_i, between the ranks for each pair of observations on *X* and *Y*, and then calculate d_i^2 (the squared difference in ranks).
3. With *n* as the sample size, the Spearman rank correlation is given as follows:

$$r_s = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)}. \qquad 3$$

Suppose an analyst is examining the relationship between returns for two investment funds, A and B, of similar risk over 35 years. She is concerned that the assumptions for the parametric correlation may not be met, so she decides to test Spearman rank correlations. Her hypotheses are H_0: $r_S = 0$ and H_a: $r_S \neq 0$. She gathers the returns, ranks the returns for each fund, and calculates the difference in ranks and the squared differences. A partial table is provided in Exhibit 4.

Exhibit 4: Differences and Squared Differences in Ranks for Fund A and Fund B over 35 Years

Year	Fund A	Fund B	Rank of A	Rank of B	*d*	d^2
1	2.453	1.382	27	31	−4	16
2	3.017	3.110	24	24	0	0
3	4.495	6.587	19	7	12	144
4	3.627	3.300	23	23	0	0
⋮						
30	2.269	0.025	28	35	−7	49
31	6.354	4.428	10	19	−9	81
32	6.793	4.165	8	20	−12	144
33	7.300	7.623	5	5	0	0
34	6.266	4.527	11	18	−7	49
35	1.257	4.704	34	16	18	324
					Sum =	2,202

The Spearman rank correlation is as follows:

$$r_s = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)} = 1 - \frac{6(2,202)}{35(1,225-1)} = 0.6916.$$

The test of hypothesis for the Spearman rank correlation depends on whether the sample is small or large ($n > 30$). For small samples, the researcher requires a specialized table of critical values, but for large samples, we can conduct a *t*-test using the test statistic in Equation 2, which is *t*-distributed with $n - 2$ degrees of freedom.

In this example, for a two-tailed test with a 5 percent significance level, the critical values for $n - 2 = 35 - 2 = 33$ degrees of freedom are ±2.0345. For the sample information in Exhibit 4, the calculated test statistic is as follows:

$$t = \frac{0.6916\sqrt{33}}{\sqrt{1-(0.6916^2)}} = 5.5005$$

Accordingly, we reject the null hypothesis (H_0: $r_S = 0$), concluding that evidence is sufficient to indicate that the correlation between the returns of Fund A and Fund B is different from zero.

EXAMPLE 3

Testing the Exchange Rate Correlation

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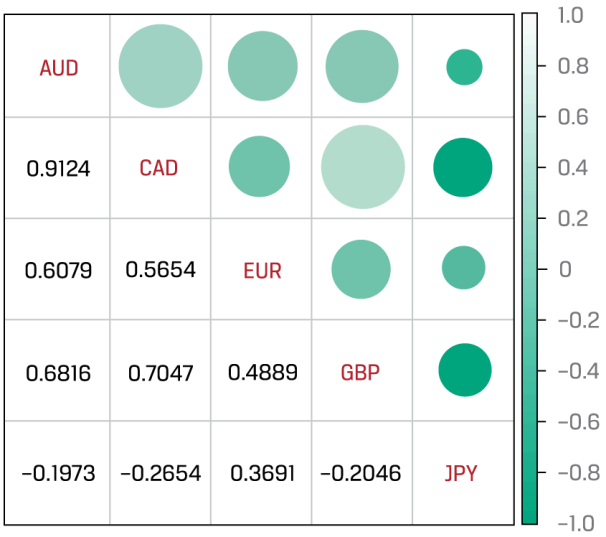
examine the relationships among these currencies using Spearman rank correlations. She calculates these correlations between the currencies over 180 days, which are shown in the correlogram in Exhibit 5. In this correlogram, the lower triangle reports the pairwise correlations and the upper triangle provides a visualization of the magnitude of the correlations, with larger circles indicating the larger absolute value of the correlations and darker circles indicating correlations that are negative.

Exhibit 5: Spearman Rank Correlations between Exchanges Rates Relative to the US Dollar



1. For any of these pairwise Spearman rank correlations, can we reject the null hypothesis of no correlation (H_0: $r_S = 0$ and H_a: $r_S \neq 0$) at the 5 percent level of significance?

Hide Solution

Solution:

The critical *t*-values for 2.5 percent in each tail of the distribution are ±1.97338.

There are five exchange rates, so there are 5C2 = {5! / [2!(5 -2)!]}, or 10, unique correlation pairs. Therefore, we need to calculate 10 *t*-statistics. For example, the correlation between EUR/USD and AUD/USD is 0.6079. The calculated *t*-statistic is

$$\frac{0.6079\sqrt{180-2}}{\sqrt{1-0.6079^2}} = \frac{8.11040}{0.79401} = 10.2144.$$

Repeating this *t*-statistic calculation for each pair of exchange rates yields the test statistics shown in Exhibit 6.

Exhibit 6: Calculated Test Statistics for Test of Spearman Rank Correlations

	AUD/USD	CAD/USD	EUR/USD	GBP/USD
CAD/USD	29.7409			
EUR/USD	10.2144	9.1455		
GBP/USD	12.4277	13.2513	7.4773	
JPY/USD	−2.6851	−3.6726	5.2985	−2.7887

The analyst should reject all 10 null hypotheses, because the calculated *t*-statistics for all exchange rate pairs fall outside the bounds of the two critical values. She should conclude that all the exchange rate pair correlations are different from zero at the 5 percent level.

Question Set

You are interested in whether excess risk-adjusted return (alpha) is correlated with mutual fund expense ratios for US large-cap growth funds. The following table presents the sample.

3	−0.50	1.90
4	−1.01	1.50
5	−0.26	1.35
6	−0.89	0.50
7	−0.42	1.00
8	−0.23	1.50
9	−0.60	1.45

1. Formulate null and alternative hypotheses consistent with the verbal description of the research goal.

Hide Solution

Solution:

We have a "not equal to" alternative hypothesis:

H_0: $\rho = 0$ versus H_a: $\rho \neq 0$

2. Identify and justify the test statistic for conducting a test of the hypotheses in Part A.

Hide Solution

Solution:

Mutual fund expense ratios are bounded from above and below; in practice, there is at least a lower bound on alpha (as any return cannot be less than −100 percent), and expense ratios cannot be negative. These variables may not be normally distributed, and the assumptions of a parametric test are not likely to be fulfilled. Thus, a nonparametric test appears to be appropriate.

We would use the nonparametric Spearman rank correlation coefficient to conduct the test:

$$r_s = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)}$$

with the t-distributed test statistic of

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}.$$

The calculation of the Spearman rank correlation coefficient is given in the following table.

Mutual Fund	Alpha	Expense Ratio	Rank by Alpha	Rank by Expense Ratio	Difference in Rank	Difference Squared
1	−0.52	1.34	6	6	0	0
2	−0.13	0.40	1	9	−8	64
3	−0.50	1.90	5	1	4	16
4	−1.01	1.50	9	2.5	6.5	42.25
5	−0.26	1.35	3	5	−2	4
6	−0.89	0.50	8	8	0	0
7	−0.42	1.00	4	7	−3	9
8	−0.23	1.50	2	2.5	−0.5	0.25
9	−0.60	1.45	7	4	3	9
						144.5

$$r_s = 1 - \frac{6(144.5)}{9(80)} = -0.20417.$$

3. Determine whether to reject the null hypothesis at the 0.05 level of significance if the critical values are ±2.306.

Hide Solution

The calculated test statistic, using the t-distributed test statistic

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \text{ is } t = \frac{-0.20417\sqrt{7}}{\sqrt{1-0.041681}} = \frac{-0.540183}{0.978937} = -0.55181.$$

On the basis of this value falling within the range of ±2.306, we fail to reject the null hypothesis that the Spearman rank correlation coefficient is zero.

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